Applied Data Mining

Mid-term project

Neutrosophic Association Rule Mining Algorithm for Big Data Analysis

Summary

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Neutrosophic Association Rule Mining Algorithm for Big Data Analysis (2018)

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In the era of big data, where the amount of data companies deal with is massive, mainly unstructured, and constantly growing, effective information processing tools turn out to be necessary. A better understanding of data leads to better decision-making. That is why Big Data Analytics has surged. Big data analytics is the often complex process of examining large and varied data sets - or big data - to uncover information including hidden patterns, unknown correlations, market trends and customer preferences that can help organisations make informed business decisions. Association rule mining is a subset of those tools. It is the data mining process of finding the rules that may govern associations and causal objects between sets of items. For example, in a given transaction with multiple items, it tries to find the rules that govern how or why such items are often bought together.
Classical association rules can be used only with binary features, and fail to deal with quantitative attributes. As a solution to this problem, fuzzy mining has surged. Fuzzy association rules can deal with the numerical attributes problem by expressing numerical values using linguistic terms. For example, ranges of temperature values can be expressed as "very cold", "cold", "warm", etc. It uses a membership-function which takes values between 0 and 1, avoiding in that way the sharp boundary problem. However, in the event that the description of an object or their relations or both is indeterminate and inconsistent, it cannot be handled by fuzzy mining. This problem led to the birth of neutrosophic association rules which handle the indeterminacy issue by using not only a membership function, but also a falsity function and an indeterminacy function.
Introducing a new neutrosophic association rule and comparing it to fuzzy mining is the purpose of this paper.

Basic concepts of association rules

Definition 0.1 (Association rule)
The problem of association rule mining is defined as follows:
Let $I = \{i_1, i_2, \ldots, i_n\}$ be a set of n binary attributes called items, and $D = \{t_1, t_2, \ldots, t_m\}$ be a set of transactions called the database. Each transaction in D has a unique transaction ID and contains a subset of the items in I.
A rule is defined as an implication of the form: $X \Rightarrow Y$, where $X, Y \subseteq I$. Every rule is composed by two different sets of items, also known as itemsets, X and Y, where X is called antecedent and Y consequent.

In order to determine which rules are relevant, constraints on various measures of significance and interest are used. The best-known constraints are minimum thresholds on support and confidence. Let X be an itemset, $X \Rightarrow Y$ an association rule and T $\subseteq$ I a set of transactions.

Support: It is an indication of how frequently the itemset appears in the data set. The support of X with respect to T is defined as the proportion of transactions t in the data set which contains the itemset X.

Confidence: It is an indication of how often the rule has been found to be true. The confidence value of a rule, X $\Rightarrow$ Y, with respect to T, is the proportion of the transactions that contains both X and Y. Confidence is defined as: $\text{conf}(X \Rightarrow Y) = \text{supp}(X \cup Y)/\text{supp}(X)$

An association rule X $\Rightarrow$ Y is addressed in D with confidence c if at least c transactions out of D contain both X and Y. The rule X $\Rightarrow$ Y is considered as a large itemset having a minimum support s if: $\sigma_{X \cup Y} \leq |D| \times s$

There are two types of association rules: ***positive association rules*** which associate events that take place together, and ***negative association rules*** which represent the relationship between the absence of an item and the presence of another.

Proposed model

The proposed model is build under the Single Valued Neutrosophic Set (SVNS) environment.

Definition 0.2 (SVNS)
Let X be a universe of discourse. A single valued neutrosophic set (SVNS) A over X is an object taking the form $A = \{\langle x, T_A(x), I_A(x), F_A(x)\rangle : x \in X\}$, *where* $T_A(x)$: $X \rightarrow [0,1]$, $I_A(x)$: $X \rightarrow [0,1]$ *and* $F_A(x)$: $X \rightarrow [0,1]$ *with* $0 \leq T_A(x) + I_A(x) + F_A(x) \leq 3$ *for all* $x \in X$. *The intervals* $T_A(x)$, $I_A(x)$ *and* $F_A(x)$ *represent the truth-membership degree, the indeterminacy-membership degree and the falsity membership degree of x to A, respectively. For convenience, a SVN number is represented by* $A = (a, b, c)$, *where* $a, b, c \in [0,1]$ *and* $a + b + c \leq 3$.

Definition 0.3 (Intersection and union)
For two SVNSs $A = \langle x, T_A(x), I_A(x), F_A(x)\rangle$ *and* $B = \langle x, T_B(x), I_B(x), F_B(x)\rangle$, *the intersection of these SVNSs is again an SVNSs which is defined as* $C = A \cap B$ *whose truth, indeterminacy and falsity membership functions are defined as* $T_C(x) = min(T_A(x), T_B(x))$, $I_A(x) = min(I_A(x), I_B(x))$ *and* $F_C(x) = max(F_A(x), F_B(x))$.
The union is defined in the same way by changing min into max, and vise versa.

The model generates rules of form: $X \Rightarrow Y$ where $X \cap Y = \varnothing$ and X,Y are neutrosophic sets.

The neutrosophic sets construction steps can be summarised as follows:

- Step 1: Input quantitative database, minimum support, minimum confidence thresholds.
- Step 2: Represent each item in quantitative database as a neutrosophic number, and determine its truth, indeterminacy, and falsity membership functions. This is done by applying the following steps to each quantitative variable:
  - Replace the quantitative variable by a linguistic one whose value is a set of n terms represented in natural language (for example, temperature can be represented by a variable which takes values in the following 3 linguistic terms: "cold", "warm", "hot").
  - Define the truth, indeterminacy, and the falsity membership functions for each constructed linguistic term (they are usually defined by experts).
  - For each transaction, compute the truth-membership, indeterminacy-membership and falsity-membership degrees. We will have as many tuples per transaction as there are linguistic terms (i.e. n).
  - For each k-terms combination where k = $\{1, 2, ...n\}$, the count is calculated by summing degrees of membership for each transaction as: $Count(A) = \sum_{i=1}^{|D|} \mu_A(x)$ where μ_A is T_A, I_A, or F_A. And the support s is calculated as follows: $s = \frac{Count(A)}{No.\ of\ \ transactions}$.
  - The variable is then represented by the linguistic term with the highest support. Its truth-membership, indeterminacy-membership, and falsity-membership functions are then set equal to those of that linguistic term.
  - The process is repeated for each variable. At the end, we obtain, for each value of k, neutrosophic sets instead of quantitative variables, and neutrosophic numbers instead of quantitative observations.

  At this stage, each transaction is represented by a set of neutrosophic numbers.
- Step 3: For each transaction calculate the count, support, and confidence.
- Step 4: If support and confidence of the transaction exceed the minimum thresholds, it is added to frequent itemsets. Otherwise, it is added to infrequent itemsets.

- Step 5: Determine all valid positive and negative association rules. Under some conditions[1], negative association rules can be determined using falsity-association and conviction rate [2].

Experiments and results

Both fuzzy mining and the proposed model have been evaluated on Telecom Egypt Company stock records. Egyptian stock market has three indicators, and the aim of the experiment is to study the relation between the stock and those three indicators.

The experiment showed that the proposed neutrosophic model generates more rules than fuzzy mining (figure 1 below).

Min-Support	0.02	0.03	0.04	0.05
1-itemset	10	10	10	10
2-itemset	37	36	36	33
3-itemset	55	29	15	10
4-itemset	32	4	2	0

No. of generated fuzzy rules

Min-Support	0.02	0.03	0.04	0.05
1-itemset	26	26	26	26
2-itemset	313	311	309	300
3-itemset	2293	2164	2030	1907
4-itemset	11,233	9689	8523	7768

No. of generated neutrosophic rules

Figure 1: Number of generated rules with different min-support thresholds

Besides positive association rules, the proposed model generates negative association rules as well, which helps increase the efficiency of decision-making processes.

However, the experiment also showed that the proposed model generates rules for most of the min-support threshold domain (figure 2 below), which may lead to huge amounts of rules requiring further mining. Hence, the min-support threshold should be chosen wisely.

Min-Support	0.5	0.6	0.7	0.8	0.9
1-itemset	11	9	9	6	5
2-itemset	50	33	30	11	10
3-itemset	122	64	50	10	10
4-itemset	175	71	45	5	5
5-itemset	151	45	21	1	1
6-itemset	88	38	8	0	0

Figure 2: No. of neutrosophic rules with different min-support thresholds

Insights

Positive rules are derived from the truth-membership function, and negative rules are derived from the falsity-membership function, while support and confidence thresholds can be determined based on the indeterminacy-membership function in an automated way. Hence, the use of truth-membership function, indeterminacy-membership function, and falsity-membership function in future models will not only help in generating negative association rules, but also in increasing the quality of the generated association rules and the relevance of chosen thresholds.

[1] $T(x) + F(x) = 1$ where T and F are truth-membership and falsity-membership functions, respectively.

[2] $conv(X \Rightarrow Y) = \frac{1-supp(Y)}{1-conf(X \Rightarrow Y)}$ can be interpreted as the frequency that the rule makes an incorrect prediction.